DA17 Merging two tables and the fan-out trap

Week 6, Messy data and two tables | Session 17 of 24 | Python for Data Analysis

Today's question. Which product category earns the most, and can we trust the merge that tells us?

Objective. By the end: you can merge two tables on a key, check the row count before and after, and spot a duplicated key that inflates a total.

> **WHERE THIS IS USED**
> Customers, products and sales usually live in separate tables, and analysts join them. A careless join can count one sale twice. How would you check that your total is still right after a join?

Before this session: DA16 Cleaning the raw export. If you cannot explain the last example, use its error card with the instructor first.

Words for today

Word	What it means	Think of it like this
merge key	A column that identifies a row and is shared by two tables, such as product_id.	The item number on both the price board and the receipt.
merge	Join two tables by matching their key columns.	Matching each receipt with its line on the price board.
left merge	Keep every row of the first table, even when nothing matches.	Keep every receipt, even for a missing price board line.
unique	Appears once only. A key must be unique in the table it looks up.	One item number per product.
fan-out	A merge that makes extra rows because a key appears twice, and inflates every total.	One receipt copied for every duplicated price line.
reconcile	Check that two routes to the same total agree, or explain why not.	Counting the cash tin and the receipts and comparing them.

Block A. A merge you can trust

sales.csv holds the product_id of each sale. products.csv says which category each product belongs to. Merge them on product_id. The row count must not change, because each sale matches exactly one product.

Predict. Predict the row count after the merge and which category earns the most. Write your guess on paper first.

Your guess, written before you run it

```
import pandas as pd
sales = pd.read_csv("sunrise_sales.csv")
products = pd.read_csv("sunrise_products.csv")
sales["line_value"] = sales["units"] * sales["unit_price_ugx"]
print(products["product_id"].is_unique)
merged = sales.merge(products[["product_id", "category"]], on="product_id", how="left")
print(len(sales), len(merged))
by_category = merged.groupby("category")["line_value"].sum().sort_values(ascending=False)
print(by_category)
print(by_category.sum())
```

Run it, then compare. The key is unique (True). Both tables have 1513 rows. The categories rank Household 2082700, Drinks 1863200, Stationery 1686500 and Snacks 743500, and they add to 6375900. The count and the total both match the true values, so the merge is safe.

Investigate.

1. Why check that product_id is unique before merging?
2. How do you know the merge did not lose or add rows?

Modify. Show the category with the lowest revenue.

Block B. The fan-out trap

sunrise_products_practice.csv is a copy of the products table with one product listed twice. Merge with it and watch the totals. No error appears.

Predict. Predict what happens to the row count and the total revenue. Write your guess on paper first.

Your guess, written before you run it

```
practice = pd.read_csv("sunrise_products_practice.csv")
print(practice["product_id"].is_unique)
bad = sales.merge(practice[["product_id", "category"]], on="product_id", how="left")
print(len(bad), bad["line_value"].sum())
print(bad["line_value"].sum() - sales["line_value"].sum())
print(practice[practice.duplicated("product_id", keep=False)])
```

Run it, then compare. The key is not unique (False). The merged table has 1628 rows, which is 115 too many, and the total is 7079900, which is 704000 too high. The duplicate is Juice with product_id 6. Python raised no error. Only the check of the row count and total finds it.

Investigate.

1. Why did Python not warn us?
2. What two numbers give the problem away?

Modify. Ask the merge to check the key by adding validate="many_to_one". Read the result.

Make

Merge the sales with sunrise_branches.csv on branch_id to get the branch names. Check that the row count is unchanged. Total the revenue by branch name and reconcile it with the totals you found in session 9.

Save your work as DA17. Keep the worked example and put your own attempt below it. Write two sentences: what you changed and how you checked it.

Error card

You see	It means	Fix
pandas.errors.MergeError: Merge keys are not unique	validate found a key that appears more than once in the table you look up.	Find the repeated key with duplicated, fix the table, and merge again.
The merged table has more rows than the sales table	A key appears twice in the second table, so rows were copied.	Compare len() before and after every merge and check the key is unique.

Exit ticket

I check that a key is unique before I merge, and I compare the row count and the total after.

AI use follows AI_USE_POLICY.pdf. In weeks 1 to 6 no AI-generated solutions are used for exercises, labs or build tasks. Practise for 20 to 30 minutes between classes by repeating the Make task with one value changed. If you are stuck for 30 minutes, save the error and bring it to class. A file saved on your laptop has not been shared until you use the route in START_HERE.pdf.